



Agentic AI in industrial engineering

Automating complex process workflows

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The automation imperative



Why traditional automation is no longer enough

The limits of rule-based systems in modern industry

Industrial processes have grown too complex and dynamic for conventional automation to handle alone. Rigid scripts and fixed decision trees break when conditions change.

- **Variability** - raw material properties, demand spikes, and equipment wear create conditions no static rule can anticipate
- **Fragmentation** - data lives across `SCADA`, `MES`, `ERP`, and IoT edge devices with no unified reasoning layer
- **Latency** - human-in-the-loop escalations introduce costly delays in time-critical operations
- **Scale** - thousands of concurrent micro-decisions per shift exceed human cognitive bandwidth

The question is no longer *whether* to automate further — it is *how* to automate decision-making itself



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What is agentic AI?



Agentic AI - beyond prompt and response

Autonomous, goal-directed systems that perceive, plan, and act

Unlike conventional LLM interactions, **agentic AI** systems operate in continuous loops - observing their environment, forming multi-step plans, invoking tools, and refining actions based on feedback.

Capability	Traditional automation	Agentic AI
Decision model	Rule-based, static	Goal-directed, adaptive
Replanning	Manual reconfiguration	Continuous, autonomous
Tool use	Fixed integrations	Dynamic tool selection
Human role	Operator	Supervisor
Error handling	Exception → halt	Retry, reroute, escalate

An agent is defined by its ability to **take actions in the world** to achieve a goal, not merely to generate text



Core architecture of an industrial AI agent

Four layers that enable autonomous process control

A well-structured industrial agent combines perception, cognition, execution, and governance into a coherent runtime.

- **Perception layer** - ingests real-time sensor streams, **OPC-UA** feeds, and historian data to build a live process state
- **Planning layer** - an LLM-powered reasoning core decomposes high-level objectives into executable subtask sequences
- **Action layer** - invokes certified tool APIs: **PLC** setpoint writes, **MES** work-order creation, supplier portal calls
- **Memory layer** - short-term context window plus long-term vector store of past incidents, procedures, and outcomes

Governance guardrails sit across all layers - enforcing safety limits, audit logging, and human-override hooks

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High-impact use cases





Predictive maintenance orchestration

From alert to resolved work order - without human dispatch

A maintenance agent monitors vibration, temperature, and oil-quality sensors across rotating assets. When anomaly patterns match a degradation model, the agent acts end-to-end.

- **Diagnoses** the fault mode by cross-referencing sensor signatures with a RAG-indexed maintenance knowledge base
- **Prioritises** the asset against production schedules and safety criticality scores
- **Creates and assigns** a `SAP PM` work order, selects the optimal technician based on skills and shift availability
- **Pre-stages** spare parts by triggering a warehouse pick request and verifying stock in `WMS`
- **Closes the loop** by updating asset health records after the technician marks the task complete

Mean time to repair reduced by **up to 40%** in pilot deployments by eliminating manual coordination steps



Autonomous quality control and defect response

Closing the detect-decide-correct cycle in real time

A vision-enabled quality agent integrates inline inspection cameras with process historians and **SPC** control charts to act on defects the moment they emerge.

- Detects out-of-spec parts using **computer vision models** calibrated per product family
- Identifies **root-cause candidates** by correlating defect patterns with upstream process parameters
- Issues automatic **setpoint corrections** to the relevant **PLC** within defined safety envelopes
- Quarantines suspect batches in **MES** and notifies quality engineers with a pre-drafted deviation report
- Learns from engineer feedback to **refine its correction heuristics** over subsequent shifts

One automotive Tier-1 supplier reduced scrap rate by **28%** within three months of agent deployment



Multi-agent coordination on the shop floor

Specialist agents collaborating to optimise the full production system

No single agent holds full context. A **multi-agent architecture** assigns specialist roles and lets agents negotiate resources, share observations, and escalate conflicts.

Agent role	Primary responsibility	Key interfaces
Scheduling agent	Dynamic job sequencing	MES , capacity model
Energy agent	Load-shifting and peak shaving	SCADA , utility API
Logistics agent	In-plant material flow	WMS , AGV fleet manager
Safety agent	Real-time hazard monitoring	SIL sensors, permit system
Orchestrator	Cross-agent conflict resolution	All agents via message bus

Agents communicate through a **shared process knowledge graph** - ensuring consistent situational awareness across the floor



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Implementation roadmap



A pragmatic path to agentic process automation

Four phases from pilot to enterprise-scale deployment

Successful industrialisation of agentic AI requires disciplined progression - proving value early while building the data and governance foundations for scale.

- **Phase 1 - Sense** (*0–3 months*) - instrument assets, unify data streams into a real-time process lakehouse, establish baseline KPIs
- **Phase 2 - Assist** (*3–6 months*) - deploy co-pilot agents that recommend actions; humans retain all execution authority
- **Phase 3 - Automate** (*6–12 months*) - grant agents write access within certified safety envelopes; expand tool inventory incrementally
- **Phase 4 - Orchestrate** (*12–24 months*) - activate multi-agent coordination; integrate with supply chain and energy market systems

Start narrow and deep - one high-value process with full observability beats ten shallow pilots with no measurable outcome



Key success factors and risks to manage

What separates production-grade agents from proof-of-concept demos

Agentic AI in industrial settings demands rigorous engineering discipline beyond model selection and prompt design.

Success factor	Risk if neglected
Certified tool APIs with hard safety limits	Agent writes unsafe setpoints to live equipment
Comprehensive audit trail for every agent action	Inability to explain or investigate incidents
Human-override at every execution step	Loss of operator trust and regulatory exposure
Continuous model drift monitoring	Silent degradation of decision quality over time
Cross-functional ownership - IT, OT, and engineering	Siloed deployments that never reach production

Industrial AI agents are **sociotechnical systems** - the organisational model matters as much as the technology stack



The autonomous factory starts with one agent

Ready to define your first use case?

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